Alex Tyh Wang

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Summary

Computer science undergraduate seeking a Spring or Summer 2027 software engineering internship in applications and engineering tools, with experience in internal platforms, production web applications, Python automation, and software testing.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, C#, SQL
Applications and Databases: Angular, React, Next.js, Node.js, ASP.NET, Spring Boot, REST APIs, PostgreSQL, SQLite
Tools: Git, Linux CLI, VS Code, Microsoft Office (Word, Excel, PowerPoint), Jenkins, CI/CD, Azure DevOps
Fundamentals: Debugging, data structures, algorithms, software testing, performance optimization, workflow automation

Experience

Software Engineer Intern – AI Integrations — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led development of an **internal AI developer platform** integrating AI agents and Model Context Protocol (MCP) tools into enterprise engineering workflows; presented the platform to executives.
- Built a custom **software evaluation harness** for large language models (LLMs), testing reliability and usefulness for developer and Sales workflows.
- Reduced token usage by **55%** with a **subagent architecture** while improving internal agent workflow performance.
- Maintained and updated **Jenkins CI/CD pipelines** supporting automated software builds and deployment.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Developed **Angular and TypeScript application interfaces** for production AI agents serving **100+ daily users**.
- Integrated OpenAI API and OpenRouter with .NET backend contributions, improving endpoint response times by **25%** and supporting **6,000+ weekly requests**.
- Reworked tool-calling and context-management software to reduce production AI token spend by roughly **50%**.

Projects

Chatio Benchmark and Chat Application — Jun 2025 – Present
Python, TypeScript, Next.js, IndexedDB, Cloudflare R2 — *chatio.dev*

- Built a ∼**500-prompt evaluation platform** combining human review, model scoring, and automated Python checks using regular expressions and JSON parsing.
- Developed a streaming web application and a persistence layer using IndexedDB and Cloudflare R2, reducing image storage by **95%** through compression.
- Published technical documentation covering evaluation design, scoring methods, limitations, and update policy.

iMessages Chatbot Server — Nov 2025 – 2026
Python, asyncio, SQLite, AppleScript — *github.com/alextyhwang/iMessages-Chatbot-Server*

- Built a **Python API gateway** integrating macOS Messages with AI services, supporting up to **20 concurrent conversations** and persistent SQLite message history.
- Implemented semaphore-controlled API calls, serialized message delivery, and debounce batching to coordinate concurrent work and target **under-2-second internal latency**.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit that taught **660+ students** through **15+ free STEM camps** and partnered with **5+ schools**.
- Led curriculum, mentor coordination, and operations, delivering **$35,000+** in free STEM education and donating **$5,000+**.
- Led communications and logistics for a **SanDisk** volunteer-day partnership building STEM kits for future programs.

Event Coordinator and Executive Board Member — Sep 2024 – Present
RIT Asian Culture Society — *Rochester, NY*

- Co-lead a **200+ member** community; help organize fundraising and Night Market, which draws **600 attendees**.